

Sharavanan Mathivanan

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EDUCATION

Western University

Bachelor of Engineering Science, Mechatronics Engineering

Bachelor of Engineering Science, Artificial Intelligence Systems Engineering

London, ON

Expected Apr 2029

Expected Apr 2029

TECHNICAL SKILLS

Programming Languages: Python, C/C++, Java, JavaScript, HTML/CSS, MATLAB, Assembly (Basic)

Robotics & Hardware: CAD Design (SolidWorks CSWA Certified, AutoCAD), PCB Layout, Soldering, Arduino, ESP32, Raspberry Pi Pico, PID Control, Sensors (IMU, Encoders, Ultrasonic), Actuators (DC/Servo)

AI & Computer Vision: TensorFlow, PyTorch, OpenCV, MediaPipe, Computer Vision

Tools & Platforms: Linux, Git/GitHub, Docker, SQL, VS Code, IntelliJ IDEA, Claude Code, MS Excel (Certified)

ENGINEERING EXPERIENCE

Payload Lead

Western Aero Design

London, ON

Sep 2025 – Present

- Lead a cross-functional team of 5, delegating across 3 sub-team leads, to build competitive UAV payload and airframe components.
- Redesigned the payload's structural mounting, wiring, and electrical interfaces based on prior-year feedback, and am leading its ongoing aircraft integration to ensure compliance and maximize scoring.
- Designed and prototyped competition UAV circuits with advanced soldering techniques, ensuring reliable avionics integration for the International SAE Challenge.

Robotics & Embedded Systems Engineer

MEBOT Edtech, Robotics, AI & IT

Chennai, India

May 2026 – Aug 2026

- Delivered a fully custom educational robot for K-12 learners, engineering its hardware, firmware, and mechanical design end-to-end from scratch.
- Expanded classroom platform support across ESP32, Raspberry Pi Pico, and Arduino R4 boards by developing a multi-language visual programming IDE for students to program robots and sprites.
- Reached 10+ students across MEBOT's programs by deploying the robot and IDE into active curriculum use.

Computer Vision and AI Application Intern

MAANTT Global Services

Chennai, India

May 2025 – Aug 2025

- Developed a real-time AI proctoring system using Python, RT-DETR, and MediaPipe for multi-person tracking, achieving 90% detection accuracy across 50+ simultaneous users.
- Automated assessment monitoring and improved user experience via eye-tracking and audio analysis features.
- Improved reliability of the proctoring engine by streamlining development workflows with GitHub and OpenCV.

PROJECTS

AMR (Autonomous Mobile Robot) v2 | *ESP32, PID Control, C++*

[GitHub](#)

- Built and validated an ESP32 PID control loop driving dual N20 motors via a TB6612FNG driver with MPU6050 gyro feedback for straight-line heading correction.
- Targeting sub-5°/m heading drift and a 100Hz loop rate, with WiFi telemetry and obstacle avoidance next toward a PCB-based, 3D-printed build.

AQM (Air Quality Monitor) Deck | *MicroPython, RP2040, IoT*

- Engineered a dual-TFT air quality monitor on a Pico WH (MicroPython), writing a hand-rolled I2C driver for BME280 sensing and a from-scratch SPI font rasterizer for the display stack.
- Integrating a PMS5003 particulate sensor and an embedded web server, targeting sub-1s dashboard load and $\pm 10\%$ accuracy across a 0-500 $\mu\text{g}/\text{m}^3$ PM2.5 range.

RT-DETR Object Detector | *PyTorch, Computer Vision*

[GitHub](#)

- Enabled real-time object detection across live video and static images by integrating RT-DETR into a PyTorch inference pipeline.
- Reduced inference latency and manual overhead by optimizing the pipeline and automating output storage.